AJ-Bench

Benchmarking Agent-as-a-Judge for Environment-Aware Evaluation

Wentao Shi, Yu Wang, Yuyang Zhao et al.
USTC · National University of Singapore · Meituan

Omer Ehrlich · AI Agent Systems, Dr. Yonatan Efroni · TAU

Nov 1, 2026

Who verifies the verifier?

Task-solving agent

Acts in an environment and produces a trajectory.

Judge

Decides whether the task really succeeded.

Task + trajectory → **Independent checks** → **Success / failure → evaluation or reward**

A credible completion message is evidence of a claim, not proof of completion.

A claim that needs fresh evidence

Claim: “The report was released on 2025-08-09.”

Text-only judge

Reads the claim and the trace.
Cannot independently confirm the date.

Agentic judge

Opens arXiv and checks the source.
Finds 2025-09-19 → rejects the claim.

Illustrative case: new evidence can overturn a plausible answer.

From judging text to checking the world

Paradigm	What it can independently check	Representative work
Rule-based	A programmed success condition	Tests / task-specific scripts
LLM-as-a-Judge	No new environment evidence	RewardBench, JudgeBench
Agent-as-a-Judge	New facts, artifacts, or state	DevAI → AJ-Bench

OSWorld / MCPMark evaluate solvers; AJ-Bench evaluates judges.

What earlier judge benchmarks established

RewardBench / RM-Bench	Judge quality across dialogue, safety and reasoning.	Lambert et al., 2025; Liu et al., 2025
	Broad content coverage; no fresh environment evidence.	
LLMBar / JudgeBench	Instruction following and reasoning correctness.	Zeng et al., 2024; Tan et al., 2025
	Stronger correctness tests; the supplied input stays static.	
AgentRewardBench	Judgment of agent task execution and planning.	Men et al., 2025
	Agent trajectories enter the evaluation; live verification does not.	

AJ-Bench asks whether the judge can acquire evidence missing from the trace.

What is new beyond earlier agentic judges?

Agent-as-a-Judge / DevAI Zhuge et al., 2025	Active verification already exists. Closest predecessor; narrower development setting.
Themis / TIR-Judge Li et al., 2024; Xu et al., 2025	Tools can support reasoning-based judgments. Less coverage of live files, database state and desktop tasks.
WebArena / OSWorld / MCPMark Zhou et al., 2023; Xie et al., 2024; Wu et al., 2025	Interactive environments for task-solving agents. They assess solvers; AJ-Bench assesses the verifying judge.

Gap: cross-domain judge evaluation with active evidence acquisition and state checks.

When does interaction improve a verdict?

MAIN RQ • SAME BASE MODEL

Text-only vs agentic: how much does the system improve?

RQ2 • INTERNAL CAPABILITY

Does more reasoning effort help?

RQ3 • EXTERNAL INPUTS

Do more interaction steps and different input formats improve judging?

Why it matters: incorrect verdicts can become misleading agent-training rewards.

Interpret the headline as a system-level comparison, not a tools-only causal effect.

Deep search versus wide search

Mind2Web2 • deep search

TASK

3 films, same 2024 theatrical date
in mainland China and the US

ONE OUTPUT ENTRY

The Beekeeper | 2024-01-12
Links for China and US

JUDGE INSPECTS

Official theatrical dates
in both countries

LABELED UNIT

One film entry

WideSearch • broad search

TASK

Eileen Gu's competition results
Jan 2015–Jan 31, 2025

ONE OUTPUT ENTRY

2019-Jan | Font Romeu
Slopestyle | silver | source

JUDGE INSPECTS

Event, discipline, podium
and linked evidence

LABELED UNIT

One table row

The output artifact is the evidence

MCPMark • FileSystem

TASK

Extract 2024 plans into timeline.txt

OUTPUT

timeline.txt: path:YYYY-MM-DD
analysis_report.md:2024-01-01

JUDGE INSPECTS

File contents against source files
Coverage, format, ordering

LABELED UNIT

Whole trajectory

OSWorld • desktop / PPT

TASK

Move image right on Slide 2

OUTPUT

Slide 2: image moved from left side → right side

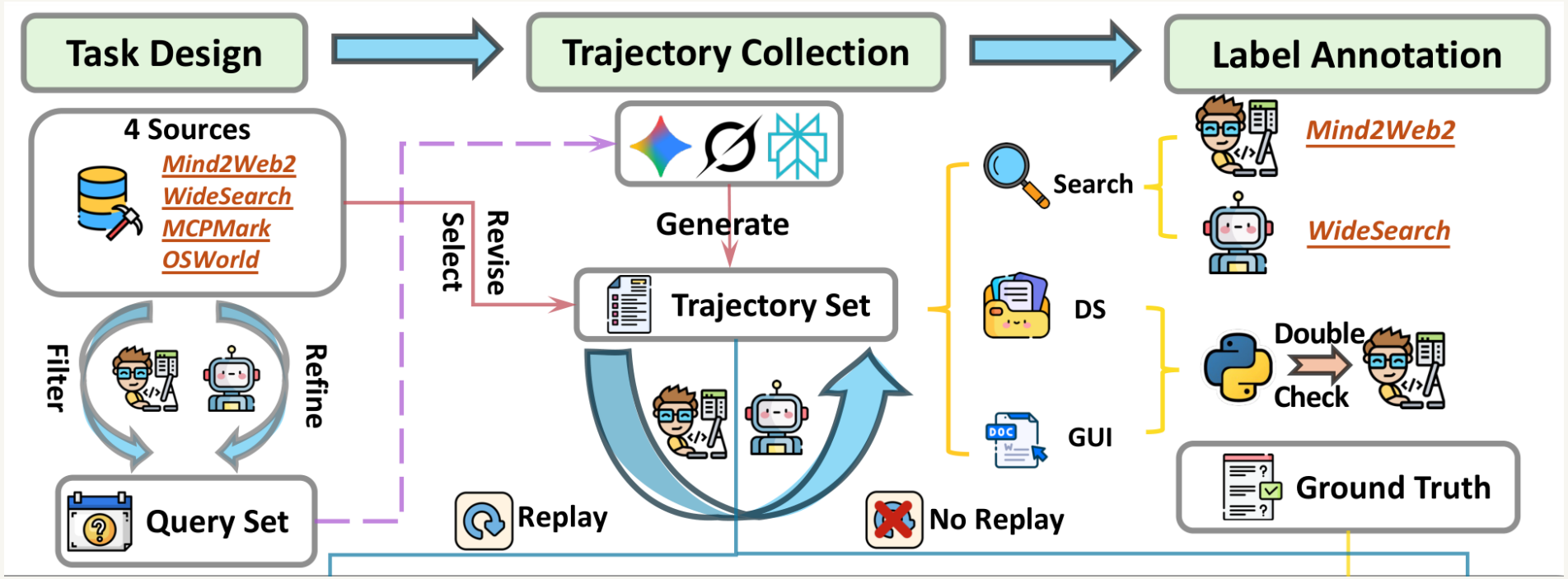
JUDGE INSPECTS

Correct slide, correct object, final position and action history

LABELED UNIT

Whole trajectory

First build the labeled benchmark



Gold labels come first.

Search: items

DS / GUI: trajectories

Fix the gold labels before the judge runs. Replay and verification follow next.

155 tasks · 516 trajectories · 60 tools

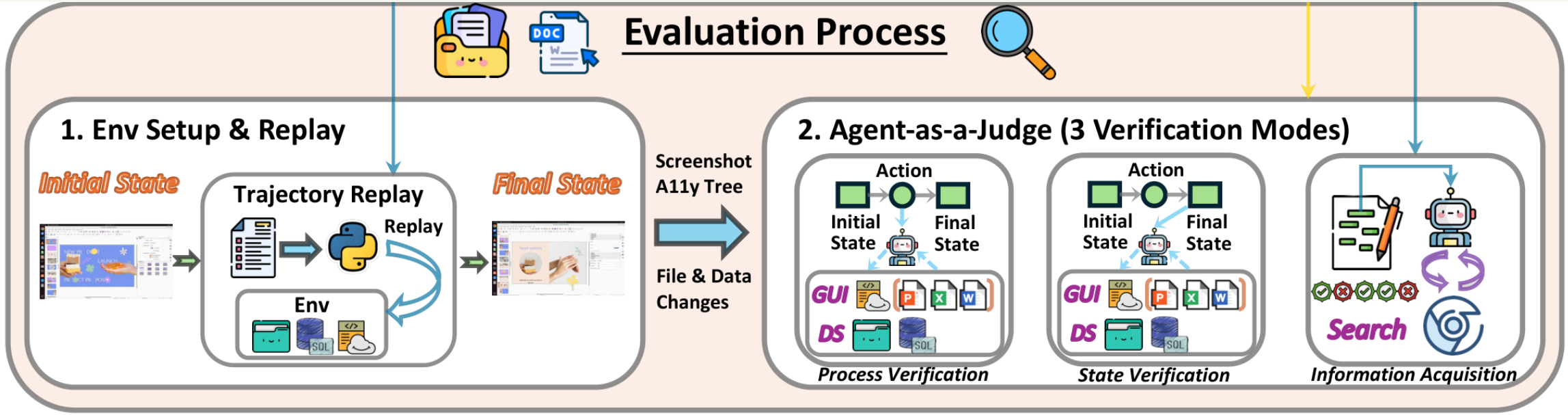
Source / domain	Subdomain	Tasks	Trajectories
WideSearch	Wide	9	27
Mind2Web2	Deep	52	156
MCPMark	FileSystem	24	129
MCPMark	Postgres	18	100
OSWorld	PPT	21	42
OSWorld	Word	12	24
OSWorld	Excel	19	38

Search:
item-level labels

File / DB / GUI:
trajectory labels

Sampling balance
≠ item balance

What changes between the two judges?



TEXT-ONLY

Task + trajectory; no live environment access

AGENTIC

Same inputs + tools, restored state and extra turns

What F1 measures, and what it misses

PRIMARY: F1

2TP / (2TP + FP + FN)

Positive class = successful work

Balances precision and recall.

Ignores true negatives; prevalence matters.

ERRORS: FALSE PASS / FALSE FAIL

FPR = FP / (FP + TN)

FNR = FN / (TP + FN)

Exposes dangerous rewards vs missed success.

A better F1 can hide a worse error type.

ADD: TASK-LEVEL UNCERTAINTY

Balanced accuracy = 1 − (FPR + FNR)/2

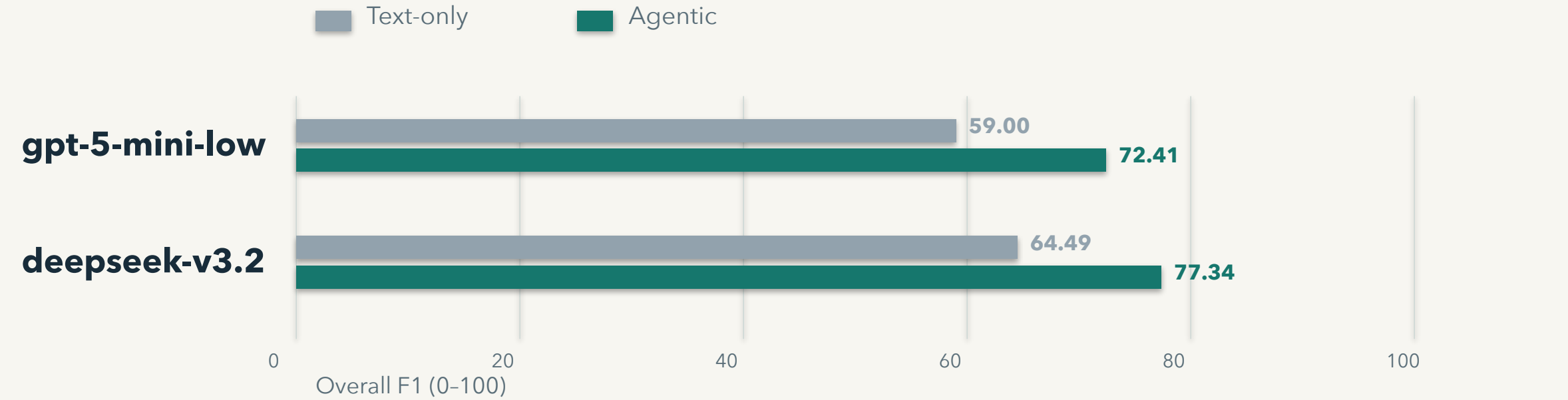
Also report dollars and p50 / p95 latency

Use task-clustered confidence intervals.

Judge value includes cost, not only agreement.

Recommendation: supplement the paper's F1 and error rates with simple baselines, cost and latency.

Agentic setup adds about 13 F1 points



Same-model gains: +13.41 / +12.85

Best text-only: Gemini 3 Pro = 75.05

Agentic deepseek is above the best text-only score; gpt-5-mini-low is not.

State inspection helps, but gains vary by model

Subdomain	gpt-5-mini-low Δ	deepseek-v3.2 Δ
Wide	+5.09	+8.82
Deep	+7.27	+19.23
FileSystem	+7.13	+12.29
Postgres	+1.78	+6.39
PPT	+31.23	+24.76
Word	+23.81	+8.87
Excel	+17.53	+9.59

Largest gpt-5-mini gains:
PPT, Word, Excel

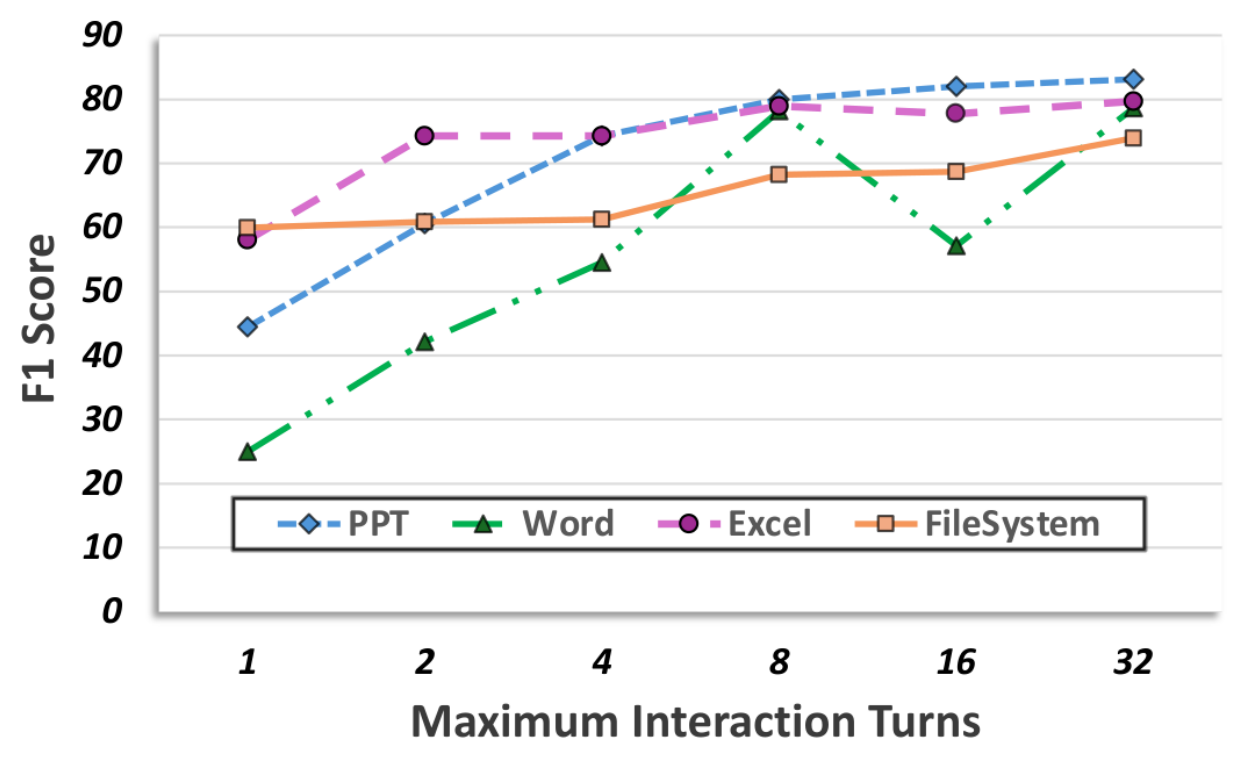
Deepseek also gains +19.23 on Deep.
No universal domain delta.

More reasoning is not reliably better

Model	Effort	Overall F1
gpt-5-mini	low	72.41
gpt-5-mini	medium	75.99
gpt-5-mini	high	73.83
deepseek-v3.2	off	77.34
deepseek-v3.2	thinking	77.11

Useful evidence > more ungrounded thought

More checks often help; not monotonically

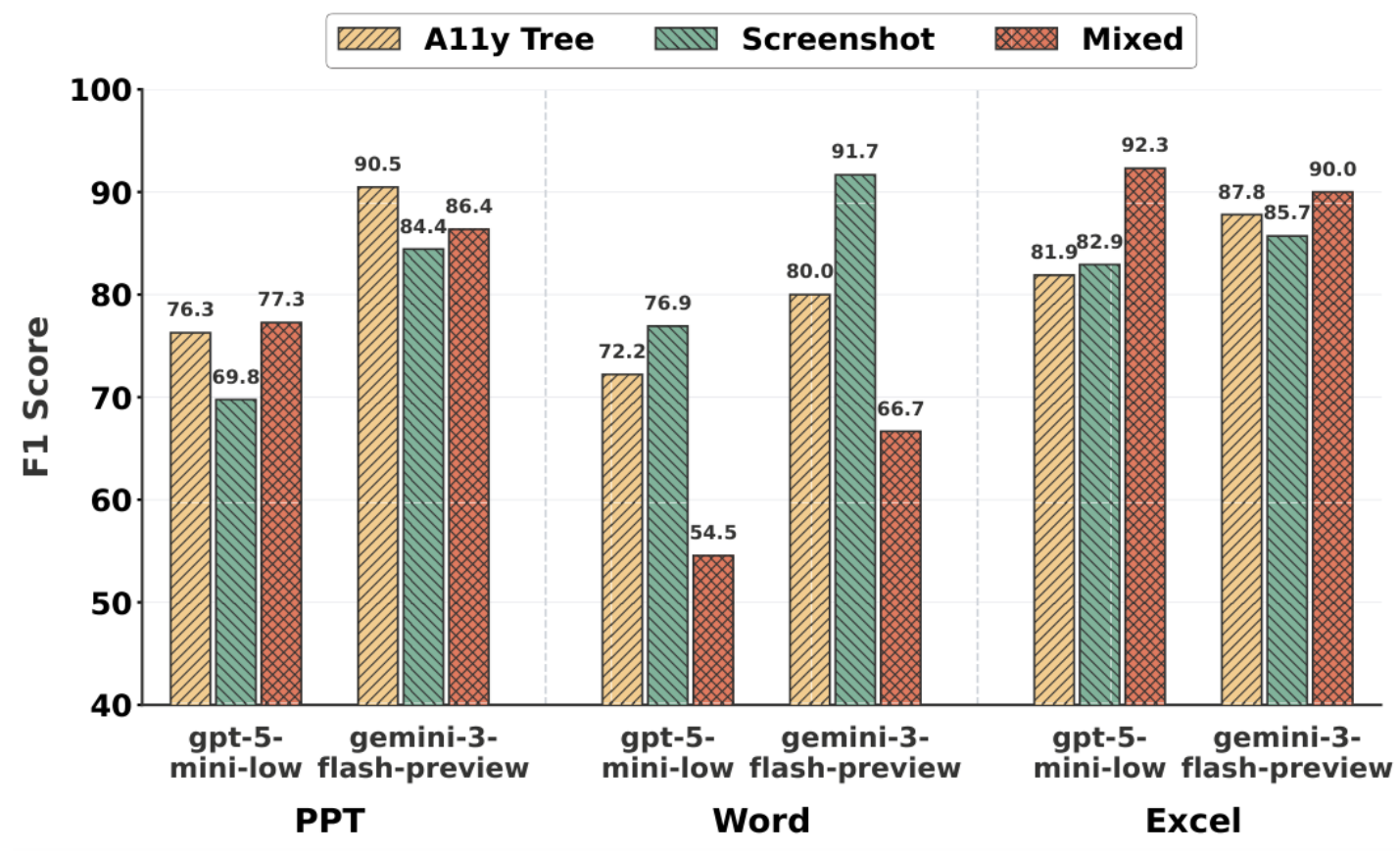


deepseek-v3.2

Big early gains from more checks

More interaction turns generally improve judging, but not every budget increase helps.

The best evidence format depends on the task



WORD: SCREENSHOT WINS

Mixed input drops F1 for both models.

PPT: model-specific winner

Excel: mixed wins for both

A11y = accessibility tree

F1 axis starts at 40.

Compare labels, not bar lengths.

What the experiments show, and what they do not

QUESTION	OBSERVED RESULT	BOUNDED INTERPRETATION
RQ1: agentic setup	Overall +13.41 / +12.85 F1 for the two full-suite models.	Supports the bundled system. Does not isolate a tools-only effect.
RQ2: reasoning	gpt-mini: medium > high; deepseek: 77.34 → 77.11.	More reasoning is not uniformly better.
RQ3: inputs / turns	Word dips at 16 turns; mixed input loses on Word.	Benefits depend on task and model.

The judge finds the right evidence, but incorrectly says PASS.

TASK

Move all duplicate-content files into duplicates/.

EVIDENCE

Judge hashes files; finds 7 duplicate pairs.
One copy is still in the root.

CRITERION

No member of any duplicate group may remain in root.

FAILURE

Judge accepts the agent's lenient interpretation → PASS.

**Found the evidence.
Understood the strict rule.
Still accepted failure.**

Most analyzed errors are interpretation / reasoning failures, not wrong-tool selection.

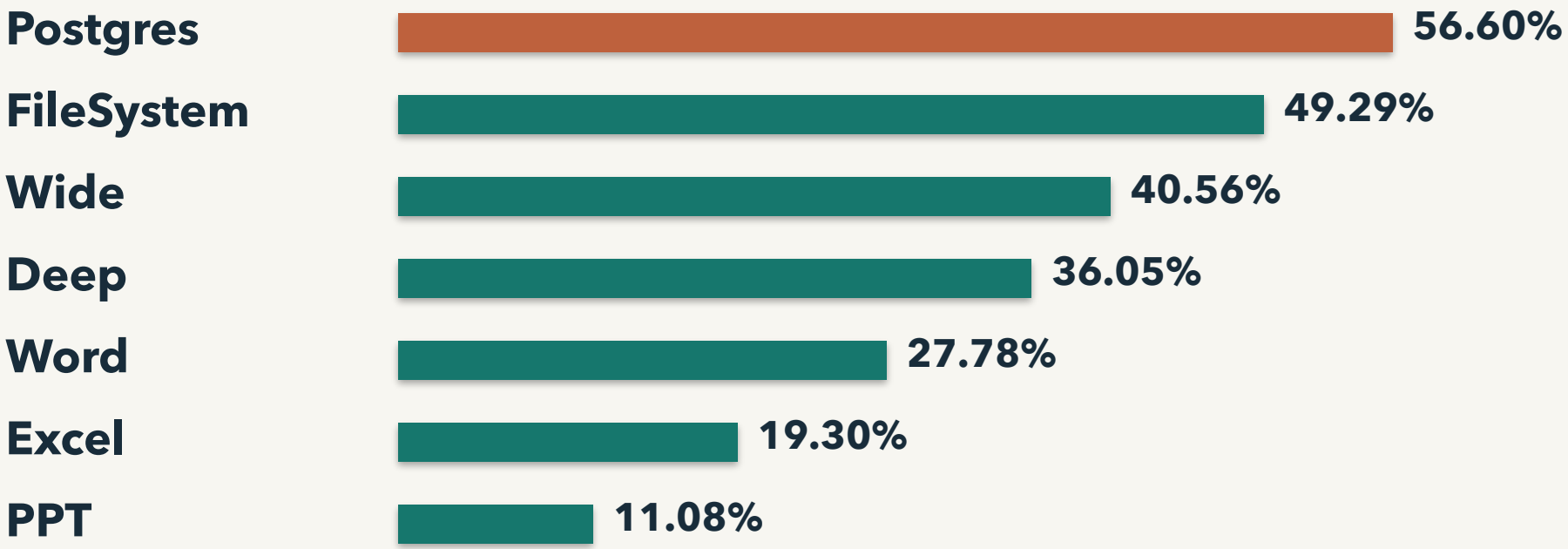
Fewer false passes can mean more missed successes

Wide judge	False-pass rate (%)	False-fail rate (%)
gpt-5-mini-low · text	39.83	36.76
gpt-5-mini-low · agent	17.65 ↓	41.06 ↑
deepseek-v3.2 · text	44.27	30.22
deepseek-v3.2 · agent	40.56 ↓	17.42 ↓

False pass: failed work gets PASS. False fail: successful work gets FAIL.

Conditional error rates on Wide • Table 9

57% of Postgres failures still receive PASS



deepseek-v3.2 agentic

Postgres FNR: 6.38%

Agentic deepseek-v3.2 false-pass rates (%) • Table 9

Independent critique: what is not established

Label provenance

WideSearch: majority vote of 6 models.
Shared-error / circularity risk remains.

Coverage and uncertainty

2 full-suite models; extra subset tests.
Task-level uncertainty not reported.

Environment validity

Replay fidelity and live-web drift.
Adapted tasks, not all built from scratch.

Operational value

End-to-end cost / latency unreported.
RL reward safety remains untested.

Fairness check: match cost before attributing the gains to tools or environment access.

Presenter critique • evidence: §3.3, Tables 3 / 6–7 and authors' Limitations

A correct verdict can hide a faulty check

OBSERVED • HISTORICAL POSTGRES DEMO

Before: max ID 59. After: 200 new rows, IDs 60–259.
Judge used current max 259 as baseline; >259 found 0.

MODEL'S CONCLUSION

“No customers were migrated” → FAIL.

THE ACTUAL DEFECT

One email was wrong. The judge never checked it.
Would it still FAIL if that email were fixed?

AJ-Bench-Causal:
Does an Agent Judge Use the Right Evidence?

1 • CHANGE DECISIVE EVIDENCE

Only customer 207's email differs.
Incorrect → FAIL; repaired → PASS.

2 • CHANGE A HARMLESS CUE

Renumber IDs consistently; preserve task.
The correct verdict should stay the same.

3 • CONTROL THE TEST

Same masked trace for the email pair.
Fix models/budgets; hold out source tasks.

4 • SCORE EVIDENCE RELIANCE

Correct flips + stable correct labels.
Audit checks, not just final answers.

Verdicts should track task correctness, not shortcuts or irrelevant cues.

The judge needs evidence, and must use it correctly

Active verification helps.
A higher F1 is not a safety guarantee.

Same-model gain: about +13 F1 points
Still unresolved: evidence interpretation and false passes

Questions: what would you require before using the judge as a reward model?

Same F1, very different false-pass behavior

Judge	TP	FP	FN	F1	False-pass rate
Always PASS	50	50	0	66.7	100%
Strict judge	25	0	25	66.7	0%

Synthetic balanced set: 50 true successes + 50 true failures. No paper attribution.

Lecture illustration • not an AJ-Bench result

Full subdomain scores for the two main models

Setting	Wide	Deep	FS	PG	PPT	Word	Excel	Overall
gpt-mini text	60.84	68.42	60.41	65.52	45.05	48.41	64.36	59.00
gpt-mini agent	65.93	75.69	67.54	67.30	76.28	72.22	81.89	72.41
deepseek text	63.65	62.91	60.31	66.31	58.38	69.77	70.12	64.49
deepseek agent	72.47	82.14	72.60	72.70	83.14	78.64	79.71	77.34

gpt-mini = gpt-5-mini-low • deepseek = deepseek-v3.2 • FS = FileSystem • PG = Postgres

Full paired main-model results • Table 3

Additional models: gains are not universal

Model	Wide: text → agent	FileSystem	PPT
Gemini 3 Flash	69.13 → 72.94	76.54 → 78.62	78.26 → 90.48
Claude Sonnet 4.5	61.02 → 69.87	69.76 → 62.72	75.61 → 78.26
Kimi K2.5	67.40 → 72.09	58.18 → 62.03	69.77 → 80.00
GLM 4.7	64.20 → 71.96	59.65 → 63.16	40.00 → 80.00

Table 6 uses a task subset within 3 subdomains, not full-suite evaluations.

Additional-model subset • Table 6; scaffold effects • Table 5

Where PaperBench fits; what running this requires

PaperBench: why graders matter

Replicate research papers against a detailed rubric.
Measures solver replication, not AJ-Bench judging.

AJ-Bench: reproduction path

Data → restore state → judge → extract verdict → score.
Search / DS / GUI have different setup costs.

Record model versions, prompts, budgets and environment snapshots. Expect live-web drift; distinguish model errors from replay and extraction failures.

Optional related-work and setup context • public AJ-Bench README